Producing U.S. Population Statistics Using Multiple Administrative Sources

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Abstract

We identify several challenges encountered when constructing U.S. administrative record-based (AR-based) population estimates for 2020. Though the AR estimates are higher than the 2020 Census at the national level, they are over 15 percent lower in 5 percent of counties, suggesting that locational accuracy can be improved. Other challenges include how to achieve comprehensive coverage, maintain consistent coverage across time, filter out nonresidents and people not alive on the reference date, uncover missing links across person and address records, and predict demographic characteristics when multiple ones are reported or when they are missing. We discuss several ways of addressing these issues, e.g., building in redundancy with more sources, linking children to their parents' addresses, and conducting additional record linkage for people without Social Security Numbers and for addresses not initially linked to the Census Bureau's Master Address File. We discuss modeling to predict lower levels of geography for people lacking those geocodes, the probability that a person is a U.S. resident on the reference date, the probability that an address is the person's residence on the reference date, and the probability a person is in each demographic characteristic category. Regression results illustrate how many of these challenges and solutions affect the AR county population estimates.

Keyword*:* administrative records, population estimates

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Introduction

Administrative records (AR) have important advantages over survey-collected data when making population estimates. Some people missed in surveys appear in AR. Our AR-based U.S. population estimate for 2020 (hereafter the 2020 AR census) is higher than the 2020 Census count by 2.3 percent (339,200,000 compared to 331,400,000), and the difference in estimates is especially large for historically undercounted populations like young children, Blacks, Hispanics, and noncitizens. It is also much cheaper to produce AR-based estimates than to conduct a survey-style census, facilitating more frequent production of AR-based estimates.

Producing AR-based estimates involves some challenges, however. To take one example, though AR population coverage is quite comprehensive at the national level, Figure 1 shows that coverage is very uneven across counties.[2] The AR census estimate is more than 15 percent lower than the 2020 Census count in 5 percent of counties. If AR people who can be linked to the 2020 Census are placed in the county where they were enumerated in the 2020 Census, only about 1 percent of counties have a more than 15 percent lower AR estimate. The share of counties with estimates that are between -1 and 1 percent different rises by 9.6 percentage points. AR estimates may thus be less accurate at lower levels of geography.

Besides locational accuracy, other challenges encountered when creating AR population estimates include person coverage completeness and its consistency across time, coverage of children in particular, distinguishing international migrants from continuous U.S. residents appearing infrequently in AR, record linkage, and the choice of demographic characteristics when multiple ones are reported or when they are missing altogether. In this paper we discuss these challenges and ways to address them.

Achieving Comprehensive Coverage

Thorough coverage is essential for making high-quality population estimates. No single U.S. AR source covers the entire population, however. Table 1 reports the percent of the 2020 AR census population covered by each individual AR source and the percent only in that source. Since some of the sources come from individual states, we also report the percentages among states included in that source. The table shows that if we were to use a single source, we could cover at most 76.4 percent of people in 2020, using Internal Revenue Service (IRS) 1040 individual tax return data. No other source comes close to covering the entire population.

To achieve more comprehensive coverage, we have combined 31 sources. Many of the sources cover a segment of the population. Three sources are for retirees (IRS 1099-R, Medicare, and Social Security Administration Master Beneficiary Record (SSA MBR)), covering about 15 percent of the population. Several sources (Department of Housing and Urban Development housing assistance programs and state welfare programs) cover low-income people who may

[2] The United States has 3,143 counties.

not need to file taxes, some cover parts of the prison population (Bureau of Prisons and U.S. Marshals Service), and others cover immigrants (e.g., U.S. Citizenship and Immigration Services (USCIS), Immigration and Customs Enforcement (ICE), and Customs and Border Protection (CBP)).

Since unevenness of AR coverage at lower levels of geography is a particular concern, we study how the challenges and potential solutions discussed in this report affect it. Table 2 reports regressions showing associations between county characteristics representing challenges/solutions and the percent difference between 2020 AR census and 2020 Census county population estimates. In the first two columns, we show regressions including one factor at a time, and the last two columns show a pooled regression. Counties where higher shares of people in the AR census come only from state welfare program (SNAP, TANF, and WIC) data tend to have higher AR census estimates. In counties where the AR noncitizen share (especially those with unknown legal status) is higher, the overall AR estimate is relatively higher than the 2020 Census. This could be evidence that the addition of sources focused on low-income people and noncitizens paid off in achieving better coverage than that achieved in survey-style data collection, or it alternatively could reflect errors in their placement by geography in the AR data. We will refer to other results from these regressions throughout the remainder of the paper.

It is especially tricky to cover children using AR. Most sources contain only adults. IRS 1040 returns include dependent children, but they only appear with about a one-year lag after birth. We have addressed this in two ways. The Census Household Composition Key (CHCK) links children to one or both parents. When a child aged 18 or under does not appear directly in any AR source near the reference date, but one or both parents do, the child is placed at the same location(s) as their mother when present, and their father otherwise. With the CHCK-provided children, the AR census estimate for children aged 0 to 17 is just 0.04 percent below the 2020 Demographic Analysis (DA) estimate, while it is 4.97 percent below DA without them (Table 3).[3] Table 2 shows that the AR census estimates tend to be higher relative to the 2020 Census in counties where more children are included through CHCK. Second, U.S.-born children under age 2 found only in the Social Security Administration Numerical Identification file (NUMIDENT) are included in the AR census in their city and state of birth. We do not include older children found only in the NUMIDENT, because most children aged 2 or above are dependents in IRS 1040 returns, and the likelihood that a child moved from their birth city increases over time. The AR census estimate for children aged 2 or under is 14.72 percent below DA before including children from the NUMIDENT and 0.97 percent below after doing so (Table 3). Table 2, though, shows a negative association between the share of children aged 0 to 1 found only in the

[3] The Census Bureau's Demographic Analysis estimates, which use comprehensive birth and death records for U.S.-born people, are a benchmark for decennial census quality and are often used when evaluating coverage of children. Jensen et al. (2020) provide methodological details.

NUMIDENT and the percent difference between the AR census and 2020 Census estimates. This could reflect low AR coverage in general in counties with more NUMIDENT adds.[4]

Despite the use of so many sources, we undoubtedly omit some U.S. residents. Inclusion of more sources could remedy this. Table 2 shows a generally positive association between the number of AR sources and AR county-level coverage relative to the 2020 Census. About 3.0 percent of the population is found only in driver's license data in the five states providing those data (Table 1). The association between the share of people in the county contributed only by driver's license data and AR coverage is positive, though only in the multivariate regression (Table 2). About 2.0 percent of the population of the 22 states providing state welfare program data are found only in those sources (Table 1), and as discussed above, contributions from these sources are positively associated with AR coverage. Having these state sources in all 50 states and the District of Columbia could thus increase coverage. We include data on federal prisons, but data on state and local prisons and jails and other types of group quarters are not currently available. Brown et al. (2023) show that many of the people enumerated in group quarters in the 2020 Census have housing unit addresses in AR, which is an issue of locational accuracy rather than coverage, but the available AR sources surely miss some of the group quarters population. Table 2 shows a negative association between the share of the county population that is in group quarters in the 2020 Census and AR coverage. Some unemployment insurance recipients may be missed by the sources in the AR census, while they are included in the National Directory of New Hires, which is not currently authorized for Census Bureau use. The homeless population is unlikely to be well covered in available AR data (e.g., they may not file taxes). Local agencies working with the homeless could potentially provide data for that group.

Another issue affecting coverage is record linkage. The Census Bureau's Person Identification Validation System (PVS) assigns a unique identifier called a Protected Identification Key (PIK) to person records that can be linked to a set of AR reference files. We only include people who have been assigned a unique identifier, because this allows us to unduplicate AR to ensure that each person is included just once in the estimates, to verify that the person is eligible to be included in the estimates, and to link in locational and demographic characteristics about them. For data of vintages before 2009, PVS covered only people with Social Security numbers (SSNs), i.e., citizens and work-eligible noncitizens. For 2009 and subsequent vintages, this was expanded to include people with Individual Taxpayer Identification Numbers (ITINs).[5] In an attempt to cover people without SSNs or ITINs in the 2020 AR census, AR with sufficient personally identifiable information (PII) to be linkable, but which did not receive a PIK, were

[4] A difference between CHCK and NUMIDENT adds that could explain the contradictory results is that CHCK adds require that the parents be found in current AR sources, while NUMIDENT adds occur when the parents are not in current AR sources.

[5] All citizens and work-eligible noncitizens can have SSNs. ITINs are nine-digit numbers in a publicly known range found in the SSN field of administrative records. They are issued by the Internal Revenue Service to people needing to pay taxes, but who are ineligible for an SSN.

unduplicated, assigned a unique identifier called an Enhanced Protected Identification Key (EPIK), and placed in a new set of reference files.[6] The PVS process was then re-run on any AR without a PIK but with sufficient PII to be linkable so that EPIKs could be assigned to individual AR. The 2020 AR census includes 6.7 million people with ITIN PIKs and 4.5 million with EPIKs, illustrating the value of expanding the PVS record linkage process. The additional linkage improves AR coverage at the local level as well, as the associations between AR ITIN and EPIK shares in the county and AR coverage are positive in Table 2.

When AR are ingested by the Census Bureau, an attempt is made to link the addresses to the Master Address File (MAF) of U.S. addresses and assign a MAFID, a unique address identifier. Not all addresses can be linked to the MAF. When initially putting together the 2020 AR census estimates, we excluded any AR without a MAFID.[7] The resulting estimates were below the Census Bureau's Population Estimates Program (PEP) estimates, and in some areas the differences were large. We then decided to add people lacking MAFIDs but who have a state of residence, increasing the population estimate by 8.7 million people.[8] The association between the share of AR people in the county without a MAFID and AR coverage is negative in Table 2, which could reflect difficulty in assigning PIKs to AR when address linkage is difficult, which we discuss next.

The inability to assign PIKs to AR can cause omissions, so comprehensive person linkage is important for coverage. Veterans Service Group of Illinois (VSGI) third-party data provide a good example of this, as 34.7 percent of the records cannot be assigned a PIK and are thus not used in the AR census. VSGI records containing SSNs are assigned PIKs at high rates. When an SSN is not present, the ability to assign a PIK varies with the specificity of the address. Post Office (P.O.) boxes may be reused by people with different residential addresses. Rural routes may not always refer to a specific housing structure. Linkage is particularly difficult when no address is available. Table 4 shows that the percent of VSGI records receiving a PIK ranges from 72.6 percent for people with blank addresses to 88.3 percent for those with street addresses, for an overall average of 87.8 percent when an SSN is present. The variation across address types is much larger for records without an SSN (2.3 percent for a blank address compared to 51.1 percent with a street address), with a significantly lower average of 48.1 percent. P.O. boxes and rural routes are more common in rural areas, leading to lower AR coverage there. Table 2 shows that a county's share of VSGI records that lack PIKs is negatively associated with AR coverage. In addition, the table shows that AR coverage is lower in counties where the U.S. Postal Service (USPS) delivers mail to a lower percentage of addresses, whether they recognize

[6] Brown et al. (2023) provide details of the procedure and show descriptive statistics about people with SSN PIKs, ITIN PIKs, and EPIKs.

[7] Every person in the 2020 Census is assigned a MAFID. If a person's address is not initially found in the MAF, follow-up activity is done to either match the address to an existing MAFID or to create a new one. We did not have the resources to do this for AR without a MAFID in the 2020 AR census project.

[8] The 8.7 million figure includes the 1.5 million children under the age of 2 found only in the NUMIDENT shown in Table 2.

the address or not. These areas are likely to be ones where P.O. box and rural route mailing addresses are more common.

One potential solution to the P.O. box linkage issue is to use a newly obtained file from USPS containing P.O. box numbers and the residential addresses of their owners, among people who obtain their P.O. box for free.[9] We plan to test the usefulness of this file.

People almost never change SSNs over their lifetime, so there should be little remaining duplication after unduplicating by SSN PIK. In contrast, a person can change ITINs, which can later be reused by others, or they can switch from using an ITIN to an SSN.[10] This could lead to some duplication in the estimates, and it could be one reason for the positive association between the AR ITIN share and relative AR coverage in Table 2.[11] Further research is needed to investigate the extent of duplication among people with ITIN PIKs and how it can be minimized.

Ongoing Census Bureau research is developing methods to accurately assign PIKs to a higher share of AR, which would improve AR population statistics coverage.

Excluding People Not Eligible to Be Counted

Though we aim to maximize coverage of people who should be included in population estimates on the reference date, we try to exclude people who are ineligible. One criterion is being alive on the reference date. We identified people who were born after April 1, 2020 (the 2020 Census reference date), using birth dates from the fourth quarter 2020 NUMIDENT. We identified people who died before April 1, 2020, using death dates from the fourth quarter 2020 NUMIDENT and information from the following administrative data sources: Bureau of Prisons, Department of Defense's Defense Manpower Data Center, IRS 1040 returns, Medicare, Selective Service System, and VSGI. In addition, we excluded people who lacked a death date in the NUMIDENT but who, based on their birth date, were 115 or older.[12] Table 5 shows the number of people excluded from the 2020 AR census because of death information from each AR source and the number and share of them who can be linked to a person in the 2020 Census. A tiny fraction of people identified as deceased in the NUMIDENT can be linked to 2020 Census records, while significant shares of those in Medicare, Bureau of Prisons, and VSGI can be linked, suggesting that more research is needed about the quality of death information coming from sources other than the NUMIDENT.

[9] These people are not provided with mail delivery to their residences and are instead given free Group E P.O. boxes.
[10] ITIN to SSN switches can happen when a person becomes eligible for an SSN.
[11] Suppose a person uses one ITIN in AR in year t-1 and a different ITIN or SSN in AR in year t. Since we use both year t-1 and t AR sources in the estimates for year t, we include the person twice.
[12] As of April 1, 2020, the oldest known person in the United States was 114. A list of the oldest people in the United States can be found at https://en.wikipedia.org/wiki/List_of_the_verified_oldest_people, viewed on July 29, 2022.

Very limited death information is available for people with ITIN PIKs or EPIKs, so erroneous inclusion of people who are not alive is more of a concern for those groups. Presence in or absence from current AR sources (signs of life) are the best information available on whether they are alive or not.

Filtering out people who could be alive, but who are not U.S. residents on the reference date, as well as people whose deaths were omitted from our death sources is discussed below.

Coverage Consistency

Consistency of coverage over time is a second important characteristic of population estimates. Otherwise, population growth estimates would be inaccurate. Some sources are not available every year. For example, the SSA MBR data are not available before 2015 or between 2016-2018, and the available years for state sources vary by state. Inconsistent availability could lead to variable coverage.

People appearing in AR sources in 2019 but not 2020 is a sign of inconsistent AR coverage. Table 2 shows that the share of such people in the 2020 AR census in the county is negatively associated with AR coverage.

The data sharing agreements for several sources used in the 2020 AR census are currently inactive.[13] Table 6 shows that the 2020 AR census total population estimate drops by 1.7 percent (5.7 million) when excluding sources no longer accessible because of inactive data sharing agreements. The declines are quite uneven across demographic groups. They are larger for people under age 45, Hispanics, non-Hispanic Asians, and non-Hispanic Some Other Race (SOR).

Some sources are not received in time to be included in estimates needed by a certain date. Two sources arrived too late to be included in the first 2020 AR census estimates that were needed by December 31, 2020. One is 2019 Medicaid data, and the other is the subset of the Tax Year 2019 IRS 1040 tax returns that had been received by IRS in 2020 but were not yet processed.[14] If they are included, the total population rises by 0.7 percent (2.4 million). The gain is larger for children and people who are neither non-Hispanic White nor non-Hispanic Two or More Races (Table 6). Having more AR people found only in 2019 Medicaid data or late Tax Year 2019 IRS 1040 returns is associated with lower county-level AR coverage when using only sources available in 2020 (Table 2).

The coverage of any given source varies across years. For example, the number of people in IRS 1040 tax returns typically changes little from year to year (Table 7). The IRS number does not

[13] All data obtained with the assistance of Executive Order 13880 were incorporated into the analytic files for the 2020 AR census by January 12, 2021.

[14] Medicaid is a government health insurance program for people with low income. Before receiving 2019 data, the most recent Medicaid data available were from 2016. We included the 2016 data in our estimates. We replaced 2016 with 2019 Medicaid data when making the estimates including late-arriving sources.

track well with Census Bureau population estimates, however, sometimes declining when the population estimates are increasing. The IRS number jumps by 10 million between 2019 (Tax Year 2018) and 2020 (Tax Year 2019), then falls by 7.8 million between 2020 (Tax Year 2019) and 2022 (Tax Year 2021), while the population estimates rise each year. The number of people with ITINs appearing in IRS 1040 returns steadily declines after 2012. The number in 2022 represents only 45.7 percent of the 2012 peak. In contrast, official estimates suggest there was little change in the size of the unauthorized immigrant population during the 2010's (Baker 2021a).[15]

One way to improve consistency is to use enough sources so that as many people as possible are covered by multiple sources. Consistent with this, Table 2 shows that the share of people with multiple AR sources is generally positively associated with AR coverage.[16] People in multiple sources are much less likely to exit the data from one year to the next for a reason other than death or to appear in the data for a reason other than birth (Table 8). For some groups the AR census has such redundancy. Though IRS 1099-R forms, SSA MBR, and Medicare data each cover 14-15 percent of the population, each source uniquely covers only <0.01, 0.04, and 0.09 percent of the population, respectively, because the three sources cover nearly the same group of people (Table 1).[17]

Distinguishing Infrequently Appearing U.S. Residents from International Migrants

An important issue to address when attempting to maximize coverage of U.S. residents and maintain consistency over time is how to treat people who do not appear in AR consistently over time or who have both U.S. and foreign addresses. We would like to include continuous U.S. residents in population estimates even if they have not appeared recently in any AR source. International migrants, however, should be excluded on reference dates when they are not U.S. residents. We use Customs and Border Protection Arrival and Departure information System (ADIS) dates of entry and exit from the United States among nonimmigrant visa holders (e.g., student and temporary work visa holders), the group most likely to enter and exit the country frequently, to exclude people who were not in the United States on the reference date.[18] Second, we require that people have a U.S. address in an AR source between January 2019 and October 2020, with a few exceptions.[19] This is not perfect, as some continuous U.S. residents may have last appeared in AR prior to 2019. And some people who are not covered by

[15] Most ITIN holders are likely to be undocumented immigrants.
[16] The people with at least one 2019 or 2020 source are included in the 2020 estimates. However, if a large fraction of the county's people in the AR census are in just one source, that may make it more likely that many people in the county are in no sources.
[17] The SSA MBR data include retirees receiving social security payments, the IRS 1099-R forms report retirement savings account distributions, and Medicare is a government health insurance program for people aged 65 or over.
[18] Brown et al. (2023) describe the rules used to exclude nonimmigrant visa holders not appearing to be U.S. residents on the reference date.
[19] The exceptions are children added from CHCK or the NUMIDENT discussed above and people in the 2016 Medicaid file.

the ADIS data that we accessed may have emigrated before the reference date despite having appeared with U.S. addresses in 2019-2020 AR sources.

It would be preferable to use ADIS data on people with all immigration statuses so that we could filter out more nonresidents.[20] ADIS alone would still not be sufficient to identify all nonresidents, though, because it misses some entry and exit activity along the U.S.-Mexico border (Baker 2021b).

We are developing models to predict the probability that a person is a U.S. resident on the reference date. The models predict presence in the decennial census, which we assume to be strong evidence that a person is a living U.S. resident on Census Day. We consider the lack of a link between a person in AR to the decennial census to be weaker evidence that a person is not a living U.S. resident, because the person could also have a decennial census record without a PIK or be a living U.S. resident who was omitted from the decennial census. One set of models is for people who have both a U.S. and foreign address in the reference year. Predictors include factors such as the number of AR sources with U.S. addresses for the person in the reference year and the number in the previous year, whether the person had a foreign address in the previous year, birthplace, citizenship, and other demographic characteristics. A second set of models is for people who appeared in AR prior to the reference year, but not during it. Explanatory variables include how long ago the person last appeared in AR, the number of sources the person appeared in during that year, whether the person had a foreign address in the meantime, birthplace, citizenship, and other demographic characteristics.[21] We plan to adjust the probabilities to take into account the possibility that the person is in the decennial census with a missing link or was erroneously omitted from it.

Achieving Locational Accuracy

Not only do we aim to include all U.S. residents in population estimates, but we also try to place them in the locations where they reside on the reference date. Some people have addresses that cannot be geocoded at lower levels of geography, while others have multiple addresses in AR. The share of people in the 2020 AR census with no MAFID (and thus no subcounty geocodes) is 2.6 percent, 51.6 percent have one MAFID, and 45.8 percent have more than one. In addition to the 8.7 million people without a MAFID, another 418,000 people have MAFIDs that lack subcounty geography in the MAF, for a total of 9.1 million without subcounty geography.[22]

MAF coverage emphasizes physical residential addresses, which are most relevant for censuses and household surveys. AR contain mailing addresses, which are not always where a person

[20] The ADIS data that we accessed contain only nonimmigrant visa holders. The data sharing agreement is inactive, so we currently have no access to ADIS data.

[21] Note that these models not only address emigration, but also deaths missed by the death information we currently use in the filtering process. Death is a reason for being omitted from current AR sources.

[22] About 300,000 of the 8.7 million people without a MAFID have only state geography.

lives. For example, a person could have a Post Office (P.O.) box mailing address in AR. The MAF contains very few P.O. boxes, however. About 59.9 percent of AR addresses lacking MAFIDs have P.O. boxes. Among AR people linked to the 2020 Census, 19.6 percent of those without AR MAFIDs are in multiunit buildings with 2 to 4 units, mobile homes, and group quarters in the 2020 Census, compared to just 8.0 percent of people with AR MAFIDs. Such structures may not have clear addresses.[23]

Unexpected characters in the address field can cause an AR address to not have a MAFID. Many AR addresses in Hawaii and Queens, New York lacking MAFIDs have dashes in their house numbers. Our project team standardized these addresses in such a way that they could be linked to the MAF, and many were assigned MAFIDs.

We are working on a method to impute tract-level geography for people without subcounty geography.[24] The model predicts whether any given potential tract the AR person could be in is the tract where they were enumerated in the 2020 Census. For each tract that an AR person could potentially be in, we construct a set of tract-level predictors, including shares of people by race/ethnicity, age group, and sex, as well as interactions between those variables and the person's own race/ethnicity, age group, and sex. We also include the shares of addresses in the tract that are of different types (e.g., street addresses, P.O. boxes, and rural routes) and interactions with the person's own address type. The rationale is that people may be more likely to live in tracts where others have similar demographics and address types to them.

To handle multiple addresses, we estimate a person-place model to predict the probability that a given address is the person's address on the reference date. Predictors include variables such as AR source indicators, the time between an administrative record's vintage and the reference date, and the number of AR addresses the person has. The test dataset is the American Community Survey (ACS), a continuously running survey of about 3.5 million U.S. households per year. The model predicts the probability that a particular AR address for a person is their address in the ACS, using the ACS tabulation month as the reference date. The coefficients are applied to all AR person-address pairs. Each person's modeled probabilities are normalized to sum to one, so that the total weight for each person in the estimates is one. We include each of the person-place pairs in the estimates, weighted by the normalized person-place probabilities. Fractions of a person may be included in multiple locations.

The average person-place probability (prior to normalization) among people with an address in the county in AR (i.e., confidence in locational placement) is positively associated with AR coverage (Table 2).

[23] We have found discrepancies in how group quarters addresses are handled between AR and the MAF. The Bureau of Prisons data provide a single address for each prison complex, while the MAF has individual entries for each building in the complex.

[24] Counties are divided into tracts, each with a population ranging from 1,200 to 8,000 people (U.S. Census Bureau (2023)).

Achieving Demographic Characteristic Accuracy

We assign age-group, sex, race, and ethnicity probabilities to each person in the 2020 AR census using a combination of AR data and previously collected decennial census and household survey data. A key challenge in compiling accurate demographic information is missing data: some people lack demographic records from AR data as well as demographic reports from census and survey data. The challenge of missing demographic data is most pronounced for people with ITIN PIKs and EPIKs, who are less likely than people with SSN PIKs to be included in our demographic data sources, and for race and ethnicity, which are missing at higher rates than age and sex even among people who are included.

Age is well measured, because the date of birth from which age is calculated is time invariant, and reported values are available for nearly everyone. For SSN PIKs, date of birth comes from the NUMIDENT. For EPIKs, it comes from the AR source where the person-place pair was found, the 2010 Census, or the ACS. Date of birth is a key variable in the PVS process, so few records without date of birth receive a PIK or EPIK. For ITIN PIKs, age comes from the same sources as it does for EPIKs, but it is missing at higher rates. We model the probability a person is in different age groups when age is missing, using first name and local area characteristics as predictors.

Sex is also well measured because it rarely changes or is missing. Data on sex comes from the same sources as data on age. The model for sex adds middle name to the characteristics used in the age model.

In contrast to sex and age values, race and ethnicity values are often missing. For SSN PIKs, race and ethnicity data come from the Census Best Race File (Ennis et al. 2018), which consolidates information from AR data, household surveys, decennial census responses, and third-party data using a set of business rules. For EPIKs, they come from the AR source where the person-place pair was found, the 2010 Census, or the ACS. This is also true for ITIN PIKs, with the exception that race and ethnicity are obtained from the Best Race File when available. The race and ethnicity models use last name and local area characteristics.

An additional challenge in assigning accurate race and ethnicity values is that reporting of these characteristics can change over time (Liebler et al. 2017). Table 9 shows the percent differences between the AR census and the 2020 Census by age group, sex, and race/ethnicity when the AR census values are constructed two different ways. The first uses previously collected values from Census Bureau decennial censuses and surveys and from AR sources, and modeled imputations for missing values. The second uses 2020 Census values for the 265,600,000 people (78.3 percent of the AR census) who can be linked between the AR census and 2020 Census. There is virtually no difference in the sex and age group results when switching to 2020 Census values, reflecting a high degree of consistency in those variables. The differences for race/ethnicity are quite large, exceeding 10 percentage point points for non-Hispanic Two or

More Races, non-Hispanic American Indian and Alaska Native, non-Hispanic Asian, and non-Hispanic Some Other Race.

There are several potential reasons for the large race and ethnicity differences with the 2020 Census for the same people. The 2020 Census questions and processing method are different than what was used previously (Jones et al. 2022), giving more opportunities to mention multiple races. Different respondents may answer another way about a person's characteristics across surveys (e.g., a parent, roommate, or neighbor may have reported about the person in the 2010 Census, while the person may have self-reported in the 2020 Census). The person may change how they view their race and ethnicity. And AR sources use fewer race categories than Census Bureau decennial census and household surveys.

We plan to test different race and ethnicity modeling approaches. For example, rather than using business rules to decide which value to use, we could estimate a latent class model, as Bycroft et al. (2023) have done for their New Zealand AR population estimates.

Conclusion

This paper discusses several challenges encountered when constructing AR-based U.S. population estimates, including how to achieve and maintain comprehensive coverage, filter out people who should not be in the estimates, place people where they reside on the reference date, and attach accurate demographic characteristics.

Our analysis suggests that the estimates benefit from using multiple sources in several ways. No one source covers everyone, so combining sources achieves more comprehensive coverage. Since a source's coverage changes over time and may not be available in all years, building in redundancy with multiple sources covering a person can improve estimate consistency over time. Having multiple sources can improve prediction about where a person resides on the reference date. If, for example, multiple sources place the person at the same address at the same time, that improves confidence that the address is their residence at the time. Building in redundancy with multiple sources mitigates the operational risk of dependency on outside data providers to continue providing data in a timely fashion.

Using multi-source data leads to choices about how to handle discrepancies across the sources. We have developed models to assign probabilities to each of the person's AR addresses. The Census Bureau has developed business rules to select demographic characteristics when they are discrepant across sources. We are considering a latent class model as an alternative to the business rules.

The effort to achieve comprehensive coverage increases the risk of erroneous inclusions. We are developing models to predict the probability that a person is a living U.S. resident on the reference date.

We find that address linkage is difficult in rural areas where the USPS does not deliver mail directly to residences. In such places there are significant discrepancies in placement of people in AR and the 2020 Census, as well as an inability to assign PIKs to some AR. The latter results in omissions of people from the AR-based estimates, contributing to lower estimates than the 2020 Census counts in those areas. This calls for more effort to link AR people in rural areas to their residential addresses.

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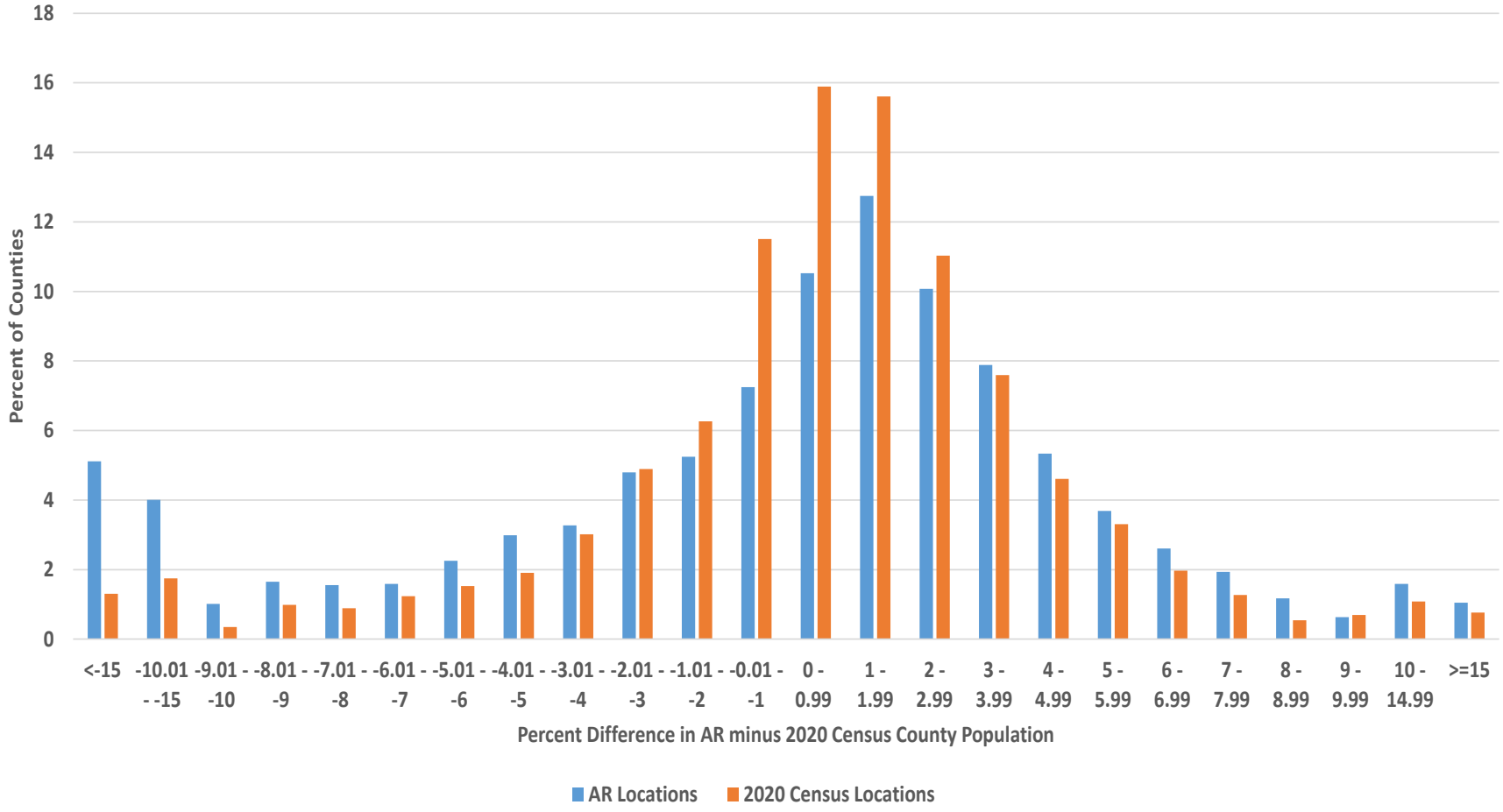
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Figure 1. Percent Difference Between AR and 2020 Census County Population, AR and 2020 Census Locations



Notes: AR location is the county where the AR data record the AR person. 2020 Census location is the county where the AR person was enumerated in the 2020 Census, for the subset of AR people who can be linked to the 2020 Census. The percent differences are calculated using the mean of the two estimates as the denominator. The data presented in this table are approved for dissemination by the DRB (CBDRB-FY23-0253).

Source: U.S. Census Bureau, P.L. 94-171 Redistricting Data and 2020 AR census.

Table 1. Percent of AR Census People Observed in Each AR Data Source

Data Source	All AR census people		AR census people in states covered by data source	
	Percent in this source	Percent only in this source	Percent in this source	Percent only in this source
Internal Revenue Service (IRS) 1040 forms	76.37	21.28	76.37	21.28
IRS 1099 forms	60.37	4.36	60.37	4.36
IRS 1099-R forms	14.08	0.00	14.08	0.00
Any IRS (1040/1099/1099-R)	88.52	36.92	88.52	36.92
Medicare Enrollment Database	15.11	0.09	15.11	0.09
Federal Housing Administration mortgage insurance	3.40	0.04	3.40	0.04
Department of Housing and Urban Development Public and Indian Housing Information Center, Tenant Rental Assistance Certification System, and Computerized Homes Underwriting Management System	0.91	0.08	0.91	0.08
Indian Health Service Patient Registration File	0.38	0.02	0.38	0.02
Social Security Administration (SSA) Master Beneficiary Record	14.99	0.04	14.99	0.04
SSA Supplemental Security Record and Special Veterans Benefits	2.56	0.14	2.56	0.14
Selective Service System	2.64	0.04	2.64	0.04
U.S. Postal Service National Change of Address	4.66	0.17	4.66	0.17
State Department passports	5.24	0.25	5.24	0.25
State Department Worldwide Refugee Admissions Processing System	0.01	0.01	0.01	0.01
Customs and Border Protection Arrival and Departure Information System (ADIS)	1.16	0.97	1.16	0.97
Immigration and Customs Enforcement (ICE) Enforcement and Removal Operations	0.24	0.19	0.24	0.19
ICE Student and Exchange Visitor Information System	0.24	0.10	0.24	0.10
U.S. Citizenship and Immigration Services (USCIS) naturalizations, lawful permanent residents, refugees, and asylees	3.59	0.19	3.59	0.19
USCIS people thought to be without lawful status on April 1, 2020	0.27	0.05	0.27	0.05
USCIS Temporary Protected Status	0.05	<0.01	0.05	<0.01
Department of Defense's (DoD) Defense Manpower Data Center	0.03	<0.01	0.03	<0.01
Department of Interior Law Enforcement Management Information System and Incident Management Analysis and Reporting System	<0.01	<0.01	<0.01	<0.01
Bureau of Prisons	0.11	0.06	0.11	0.06
U.S. Marshals Service	0.02	0.02	0.02	0.02
Veterans Service Group of Illinois (VSGI)	41.62	4.86	41.62	4.86

Data Source	All AR census people		AR census people in states covered by data source	
	Percent in this source	Percent only in this source	Percent in this source	Percent only in this source
Alaska Permanent Fund Division (state source)	0.12	<0.01	58.10	0.35
State driver's licenses	2.84	0.15	58.41	3.03
Supplemental Nutrition Assistance Program, Temporary Assistance for Needy Family, and Women, Infants, and Children programs (state sources)	3.76	0.85	8.82	2.00
Census Household Composition Key	1.06	1.06	1.06	1.06
2016 Medicaid	0.33	0.33	0.33	0.33
SSA Numerical Identification file (NUMIDENT) (ages 0-1)	0.43	0.43	0.43	0.43

Notes: The denominator for the percentages in the first two columns is the total number of people in the AR census, and the denominator in last two columns is the number of people in the AR census in states covered by the particular source. The data presented in this table are approved for dissemination by the DRB (CBDRB-FY23-0253).

Source: U.S. Census Bureau, 2020 AR census.

Table 2. OLS Regression Estimates of Associations Between County Characteristics and the Percent Difference Between the AR Census and 2020 Census County Population Estimates

	Regressions with variables representing single county characteristic		Multivariate regression	
	Coefficient	Standard error	Coefficient	Standard error
SNAP, TANF, and WIC	0.614	0.147	1.063	0.114
Naturalized citizens	0.308	0.203	-0.296	0.058
Legal noncitizens	-0.225	0.267	0.715	0.051
Noncitizens with unknown legal status	0.741	0.158	0.765	0.049
CHCK add	2.004	0.147	0.813	0.110
NUMIDENT ages 0 to 1 add	-1.391	0.323	-0.566	0.140
Driver's license	-1.075	0.148	0.683	0.139
2020 Census group quarters population	-0.950	0.055		
AR EPIK	0.654	0.169		
AR ITIN	0.669	0.124		
AR lacks MAFID	-0.520	0.050	-0.399	0.034
VSGI records without PIK	-0.383	0.040		
USPS does not recognize address	-0.184	0.013	-0.116	0.014
USPS commercial address	3.053	0.740	0.391	0.396
USPS does not deliver mail to address	-0.029	0.025	-0.141	0.014
No vintage-2020 AR source	-0.254	0.071	-0.144	0.041
2019 Medicaid	-2.922	0.346	-1.105	0.104
Late 2019 IRS 1040	-1.701	0.463		
Two AR sources	0.312	0.158	-0.126	0.055
Three AR sources	0.266	0.056	0.286	0.044
Four AR sources	0.654	0.126	0.130	0.052

	Regressions with variables representing single county characteristic		Multivariate regression	
	Coefficient	Standard error	Coefficient	Standard error
Five or more AR sources	-0.062	0.076	-0.161	0.037
Mean person-place probability	0.276	0.050		

Notes: These are OLS regressions with a dependent variable of the percent difference between the 2020 AR census and 2020 Census county population estimates, calculated using the mean of the two estimates as the denominator. The first two columns of regression results come from regressions run separately for each explanatory variable or group of explanatory variables, separated by a blank row. Specification selection for the multivariate regression was performed using a backward stepwise procedure, which dropped some of the variables. The standard errors are robust. Variable definitions are in Appendix Table 1. This table does not show the results for some of the variables in the multivariate regression. The complete results are in Tables 70 and 71 of Brown et al. (2023). The data presented in this table are approved for dissemination by the DRB (CBDRB-FY23-0253).

Source: U.S. Census Bureau, P.L. 94-171 Redistricting Data and 2020 AR census.

Table 3. AR Census and Demographic Analysis Child Population Estimates

	Number of children	Percent difference with 2020 Demographic Analysis
	Age 0-2	
2020 Demographic Analysis middle estimates	11,420,000	0.00
2020 AR census without age 0-1 additions from NUMIDENT	9,854,000	-14.72
2020 AR census with age 0-1 additions from NUMIDENT	11,310,000	-0.97
	Age 0-17	
2020 Demographic Analysis middle estimates	74,660,000	0.00
2020 AR census without CHCK	71,040,000	-4.97
2020 AR census with CHCK	74,630,000	-0.04

Notes: The 2020 Demographic Analysis middle estimates are from Jensen et al. (2020). The percent differences are calculated using the mean of the two estimates as the denominator. The data presented in this table are approved for dissemination by the DRB (CBDRB-FY23-0253).

Source: U.S. Census Bureau, 2020 Census Edited File, 2020 AR census, 2020 Demographic Analysis, 2020 Population Estimates Program, and 2020 Post-Enumeration Survey.

Table 4. Percent of Third-Party Person Records Without SSNs Receiving a PIK by Address Type

Address type	Percent of records with SSN receiving a PIK	Percent of records without SSN receiving a PIK
Street address	88.26	51.12
Post Office box	86.39	33.40
Rural route	80.96	9.21
Blank street address	72.64	2.27
Total	87.79	48.05

Notes: These percentages are among records that do not contain an SSN. The data presented in this table are approved for dissemination by the DRB (CBDRB-FY23-0253).

Source: Veterans Service Group of Illinois (VSGI) third-party records, July 2020.

Table 5. Source Contributions to Identification of Deceased Person Records

Source	This source's number of AR census people dropped because deceased	This source's number of AR census people dropped because deceased, linked to 2020 Census person	Percent linked to 2020 Census people
NUMIDENT	52,500,000	183,700	0.35
IRS 1040	404,000	365,900	90.57
Medicare Death	4,300	821	19.09
Medicare Part A Termination	94,000	72,290	76.90
Medicare Part B Termination	1,206,000	829,300	68.76
Medicare Parts A & B Termination	659,000	473,200	71.81
SSS	80	(D)	(D)
DMDC; IMARS	(D)	(D)	(D)
BOP	350	75	21.43
VSGI	72,500	60,850	83.93
Age over 114	19,000	324	1.71

Notes: "(D)" signifies that the cell is suppressed because of disclosure avoidance. The data presented in this table are approved for dissemination by the DRB (CBDRB-FY23-0253).

Source: U.S. Census Bureau, 2020 Census Unedited File and 2020 AR census.

Table 6. Percent Difference in Population Estimates Without No Longer Accessible Sources and With Late-Arriving Sources

	Percent difference in population when dropping sources no longer accessible	Percent difference in population when adding late-arriving sources
Total population	-1.69	0.71
Male	-1.73	0.77
Female	-1.66	0.64
Age 0-2	-2.24	1.06
Age 3-5	-1.71	0.88
Age 6-14	-1.88	0.92
Age 15-17	-2.36	0.93
Age 18-24	-2.81	0.48
Age 25-34	-2.39	0.53
Age 35-44	-1.87	0.70
Age 45-54	-1.34	0.67
Age 55-64	-0.97	0.74
Age 65-74	-0.87	0.62
Age 75 and over	-0.67	0.76
Hispanic	-4.80	1.08
Non-Hispanic American Indian and Alaska Native	-0.89	2.76
Non-Hispanic Asian	-4.47	1.22
Non-Hispanic Black	-0.75	1.00
Non-Hispanic Native Hawaiian and Pacific Islander	-0.78	1.54
Non-Hispanic White	-0.51	0.41
Non-Hispanic Some Other Race	-4.89	1.06
Non-Hispanic Two or More Races	-1.16	0.45

Notes: The data presented in this table are approved for dissemination by the DRB (CBDRB-FY23-0253).
Source: U.S. Census Bureau, 2020 AR census.

Table 7. People in IRS 1040 Returns Compared to Census Bureau Population Estimates

Year	People in IRS 1040's	ITINs in IRS 1040's	Vintage 2019 Population Estimates	Vintage 2022 Population Estimates
2010	272,500,000	8,527,000	309,300,000	N.A.
2011	275,600,000	9,109,000	311,600,000	N.A.
2012	277,600,000	9,480,000	313,800,000	N.A.
2013	277,000,000	8,924,000	316,000,000	N.A.
2014	278,200,000	8,491,000	318,300,000	N.A.
2015	278,900,000	8,145,000	320,600,000	N.A.
2016	280,900,000	7,848,000	322,900,000	N.A.
2017	279,800,000	7,097,000	325,000,000	N.A.
2018	281,600,000	6,465,000	326,700,000	N.A.
2019	281,500,000	5,160,000	328,200,000	N.A.
2020	291,500,000	4,481,000	329,900,000	331,500,000
2021	289,500,000	4,559,000	N.A.	332,000,000
2022	283,700,000	4,334,000	N.A.	333,300,000

Notes: The data presented in this table are approved for dissemination by the DRB (CBDRB-FY23-014-054).
Source: IRS 1040 returns; U.S. Census Bureau, vintage-2019 and vintage-2022 Population Estimates Program estimates.

Table 8. Percent Distribution of Number of AR Sources for People in Both 2020 and 2021 Estimates Compared to Others

Number of sources	In 2020 and 2021 data	In 2020, not in 2021, not death	Not in 2020, in 2021, not birth
1	20.34	88.71	70.86
2	42.47	8.53	14.95
3	25.84	1.75	8.26
4	7.79	0.64	2.85
5	2.48	0.25	1.43
6	0.73	0.09	1.02
7	0.30	0.03	0.57
8	0.04	0.01	0.05
9	0.01	<0.01	0.01
10	<0.01	(D)	<0.01
11	<0.01	0.00	(D)
12	(D)	0.00	0.00

Notes: "(D)" signifies that the cell is suppressed because of disclosure avoidance. The data presented in this table are approved for dissemination by the DRB (CBDRB-FY23-014-054).
Source: U.S. Census Bureau, administrative records data for 2020 and 2021 population estimates.

Table 9. Percent Difference Between 2020 AR Census and 2020 Census by Sex, Age, Race, and Ethnicity

	2020 AR census using previously collected values and imputations	2020 AR census using 2020 Census values for linked people	Percentage point difference between using 2020 Census values and previously collected/ imputations
Male	3.74	3.74	0.00
Female	0.94	0.94	0.00
Age 0-2	5.64	5.64	0.00
Age 3-5	5.56	5.40	-0.16
Age 6-14	0.82	0.71	-0.11
Age 15-17	-0.62	-0.70	-0.08
Age 18-24	-0.71	-0.64	0.06
Age 25-34	5.70	5.70	0.00
Age 35-44	4.68	4.77	0.09
Age 45-54	4.87	4.92	0.05
Age 55-64	2.34	2.32	-0.02
Age 65-74	-2.79	-2.79	0.00
Age 75+	-1.38	-1.38	0.00
Hispanic	14.96	10.77	-4.19
Non-Hispanic American Indian and Alaska Native	54.43	17.50	-36.93
Non-Hispanic Asian	-18.91	-2.22	16.69
Non-Hispanic Black	6.61	4.67	-1.94
Non-Hispanic Native Hawaiian and Pacific Islander	3.79	2.38	-1.41
Non-Hispanic White	1.76	-0.37	-2.13
Non-Hispanic Some Other Race	24.28	11.86	-12.42
Non-Hispanic Two or More Races	-71.42	-4.84	66.58

Notes: The percent differences are calculated using the mean of the two estimates as the denominator. The percentage point differences in column three are calculated as the column two value minus the column one value. The data presented in this table are approved for dissemination by the DRB (CBDRB-FY23-0253).

Source: U.S. Census Bureau, 2020 Census Demographic and Housing Characteristics file and 2020 AR census.

Appendix Table 1. Definitions of County-Level Explanatory Variables

Explanatory variable	Definition
Percent difference	100 * (AR census county population – 2020 Census county population)/[(AR census county population + 2020 Census county population)/2]
SNAP, TANF, and WIC	The percent of AR census people in the county who are found only in SNAP, TANF, or WIC data.
Naturalized citizens	The percentage of the county's 2020 AR census people who are naturalized citizens.
Legal noncitizens	The percentage of the county's 2020 AR census people who are legal noncitizens.
Noncitizens with unknown legal status	The percentage of the county's 2020 AR census people who are noncitizens with unknown legal status.
CHCK add	The percent of AR census people in the county who have no AR source with an address, but who are found in CHCK.
NUMIDENT ages 0 to 1 add	The percent of AR census people in the county who are in the NUMIDENT, but no other AR source, and who were under the age of 2 on April 1, 2020.
Driver's license	The percent of AR census people in the county who are found only in driver's license data.
2020 Census group quarters population	The percent of 2020 Census people in the county who were enumerated in group quarters.
AR EPIK	The percent of AR census people in the county who have an EPIK.
AR ITIN	The percent of AR census people in the county who have an ITIN.
AR lacks MAFID	The percent of AR census people in the county who do not have a MAFID.
VSGI records without PIK	The percent of person records in VSGI with an address in the county that do not have a PIK.
USPS does not recognize address	The percent of addresses in the county that are not in the USPS Spring 2020 Delivery Sequence File. The omitted category for does not recognize address, commercial, and does not deliver mail to address is residential address with USPS delivery.
USPS commercial address	The percent of addresses in the county that are classified as commercial in the USPS Spring 2020 Delivery Sequence File.
USPS does not deliver mail to address	The percent of addresses in the county that are recognized by the USPS, but do not receive mail delivery according to the USPS Spring 2020 Delivery Sequence File.
No vintage-2020 AR source	The percent of AR census people in the county who have no AR sources from 2020.
2019 Medicaid	Here 2016 Medicaid records are dropped from the AR census, and 2019 Medicaid records are added. This is the percent of AR census people in the county who are found only in 2019 Medicaid data.
Late 2019 IRS 1040	The percent of people in the county who are in a late 2019 IRS 1040 return with an address in the county. A late return is one that was not delivered to the Census Bureau until 2021. The denominator is the number of people in the AR census in the county after adding the late 2019 IRS 1040 returns. People already in the AR census are placed only in their late 2019 IRS county when calculating this variable.
Two (three, four, five or more) AR sources	The percent of AR census people in the county who have two (three, four, five or more) AR sources. The omitted category for number of AR sources is one AR source.

Explanatory variable	Definition
Mean person-place probability	The mean of the person-place probabilities from the random forest model among all person-address records in the AR census in the county.